Project Report

BUAN 6341.0W1: Applied Machine Learning
Group 6: Arbazz Alif, Prashanth Reddy, Athithyaa Muthuraman, Yamini Kandan, Opeyemi Basiru, Pranav Posina

1. MOTIVATION

Rising healthcare costs in the U.S. have increased the need for accurate, data-driven cost prediction to support financial planning and risk management. With 100,000 records and 54 features, our dataset captures the demographic, lifestyle, and clinical factors that influence insurance spending. Reliable cost prediction helps insurers optimize premium pricing, improves customer segmentation, and reduces financial uncertainty. Machine learning provides a competitive advantage by uncovering complex patterns that traditional actuarial methods may overlook. Our objective is to build robust models that estimate annual medical cost using key patient attributes. Preprocessing and feature engineering enhance model accuracy by addressing skewed healthcare cost data. SHAP analysis strengthens business transparency by explaining the primary drivers behind predicted medical expenses.

2. DATA

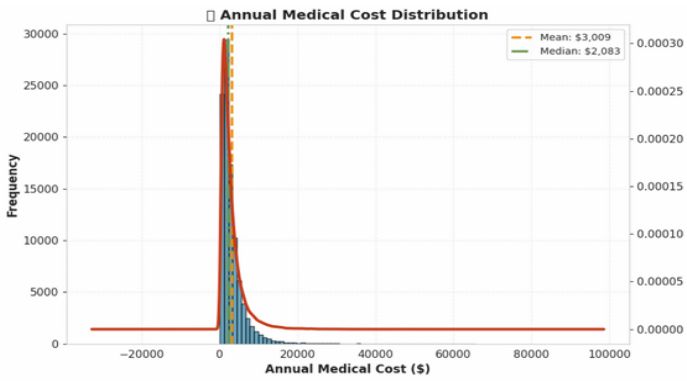
2.1 Dataset Description

- The dataset contains 100,000 observations and 54 features, capturing a wide mix of demographic, lifestyle, clinical, financial, and utilization attributes.
- Features are grouped into six major categories:
  - Demographics & Socioeconomic - 11 features
  - Lifestyle & Habits - 6 features
  - Health & Clinical Indicators - 16 features
  - Healthcare Utilization - 10 features
  - Insurance & Policy Details - 7 features
  - Medical Cost & Claims Information - 6 features
- The target variable is annual_medical_cost, representing each individual's total yearly medical expenditure.
- This diverse feature structure enables comprehensive modeling of the factors influencing healthcare costs and supports robust regression-based prediction.

2.2 Exploratory Data Analysis (EDA)

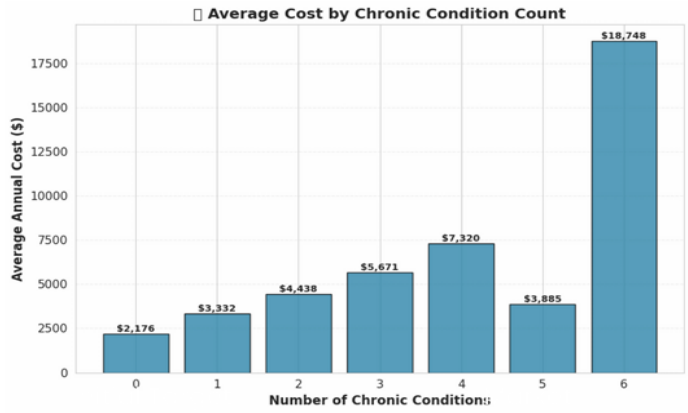
The dataset shows clear variation in medical costs driven by lifestyle, clinical history, insurance attributes, and healthcare utilization.

Initial EDA highlights strong skewness in the target variable and reveals several feature groups such as age, chronic conditions, and service usage that significantly influence annual medical cost.



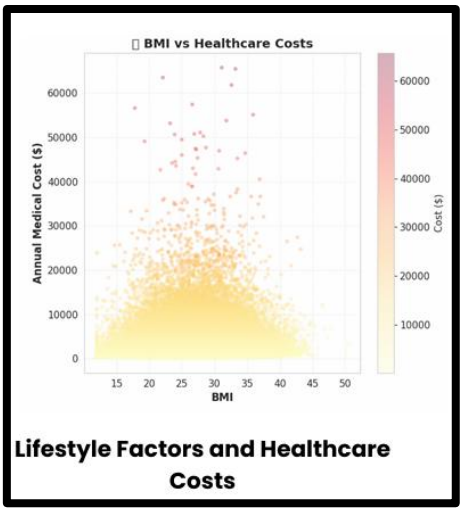
Annual Medical Cost Distribution – Understanding Our Target

- The target variable is strongly right skewed, with most patients having low costs and a small group incurring extremely high expenses.
- The means exceeds the median, showing the influence of high-cost outliers and the need for transformations like log-scaling.
- This variation indicates that medical costs differ widely across individuals, driven by health status, lifestyle, and utilization patterns.

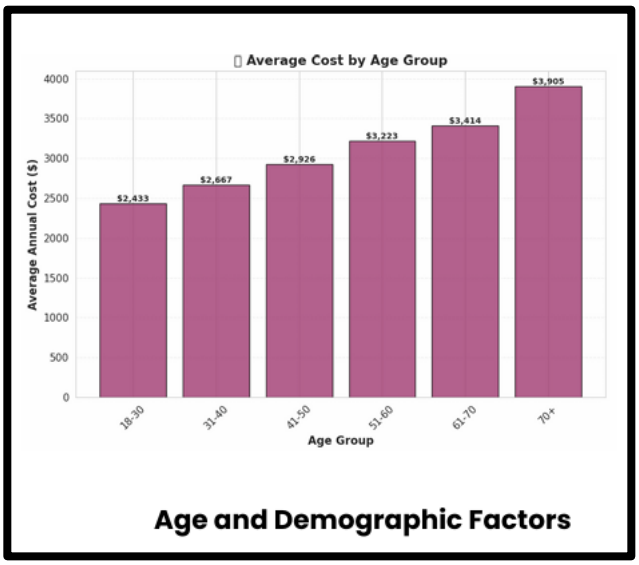


Chronic Conditions and Their Impact on Costs

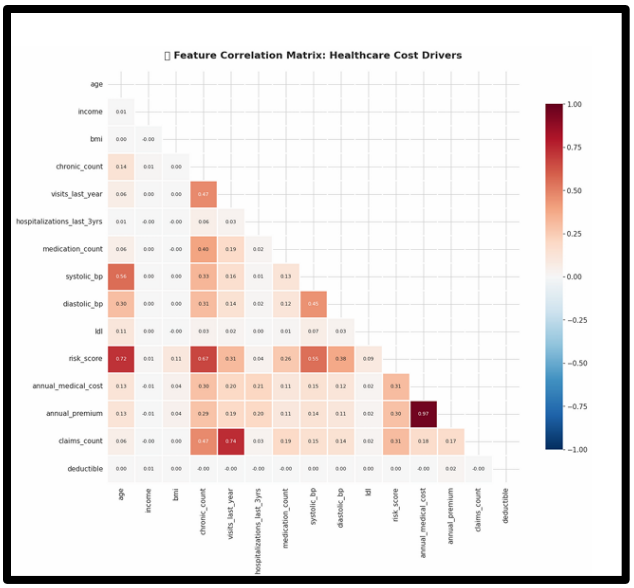
- Medical costs increase steadily with the number of chronic conditions, showing a strong positive relationship between disease burden and annual expenses.
- Individuals with six chronic conditions experience a dramatic spike in cost, reaching over $18,000, indicating significantly higher healthcare utilization.
- Conditions such as diabetes, hypertension, and liver disease are among the costliest drivers, reinforcing their importance in predicting medical expenditures.



- The scatter plot shows that higher BMI values are generally associated with increased annual medical costs, indicating BMI's role as a significant lifestyle-related predictor.
- While costs vary across all BMI levels, individuals in the overweight and obese range tend to appear more frequently among the higher-cost cases.
- The wide spread of points suggests that BMI interacts with other health and clinical factors, reinforcing the importance of including lifestyle variables in cost prediction models.



- Average annual medical cost increases steadily across age groups, showing a strong positive relationship between age and healthcare spending.
- Individuals aged 70+ incur the highest costs, reflecting greater healthcare needs and chronic condition prevalence in older populations.
- This trend confirms age as one of the most influential demographic predictors in medical cost modeling.



- **Annual premium**, **risk score**, and **chronic condition count** show the strongest positive correlations with annual medical cost, indicating that financial and clinical risk factors are major cost drivers.
- **Healthcare utilization variables -** such as visits and hospitalizations over the last three years also correlate strongly with higher expenses, reinforcing their predictive importance.
- **Clinical measures** (BP, LDL, HbA1c) exhibit moderate relationships with cost, while income and deductibles show very weak negative correlations, suggesting minimal impact on overall spending.

2.3 Data Cleaning

- A missing value check revealed 30,083 missing entries in the alcohol_freq feature, which were imputed as "0" (non-drinker), since the pattern indicated that blank entries represented individuals who do not consume alcohol.
- The column person_id was removed because it is an identifier and carries no predictive value for modeling.
- A duplicate check showed 0 duplicated records, confirming data consistency and minimizing the need for further deduplication.
- Overall, these cleaning steps ensured that the dataset was complete, consistent, and ready for preprocessing and feature engineering.

2.4 Data Preprocessing

- **Encoding:** Machine learning models require numerical inputs, so all categorical variables were converted into numeric form using LabelEncoder. This transformation allowed the models to interpret and learn from non-numeric attributes such as age_group, gender, region, smoking_status, alcohol_freq, and coverage_type. Encoding ensured consistent representation of categorical information and prevented issues arising from text-based values.

- **Scaling:** Feature scaling was applied selectively using **StandardScaler**, but only for models that are sensitive to feature magnitude - specifically Ridge Regression and the Neural Network. Scaling standardized the relevant variables to a mean of 0 and a standard deviation of 1, ensuring that no single feature dominated model learning due to its numeric range. This step improved convergence and stability for magnitude-dependent algorithms.

As a final preprocessing step, the dataset was split into 80% training and 20% testing data to support effective model training and performance evaluation.

3. FEATURE ENGINEERING

Feature engineering was performed to enhance signal quality, reduce redundancy, and improve model interpretability and performance. The process included creating new high-value predictors, removing irrelevant or duplicated features, and selecting the most impactful variables.

3.1 Feature Creation

To improve model performance, we created a small set of meaningful features that captured important behavioral and clinical patterns without introducing data leakage.

- **age_group** was derived by categorizing individuals into Young, Adult, Middle, and Senior groups, helping models better capture nonlinear age effects.
- **claim_load_ratio** was engineered as

$$claim_load_ratio = \frac{total_claims_paid}{annual_premium + 1}$$

This feature quantifies how heavily an individual utilizes their insurance relative to the premium paid, making it a strong proxy for financial risk.

3.2 Feature Reduction

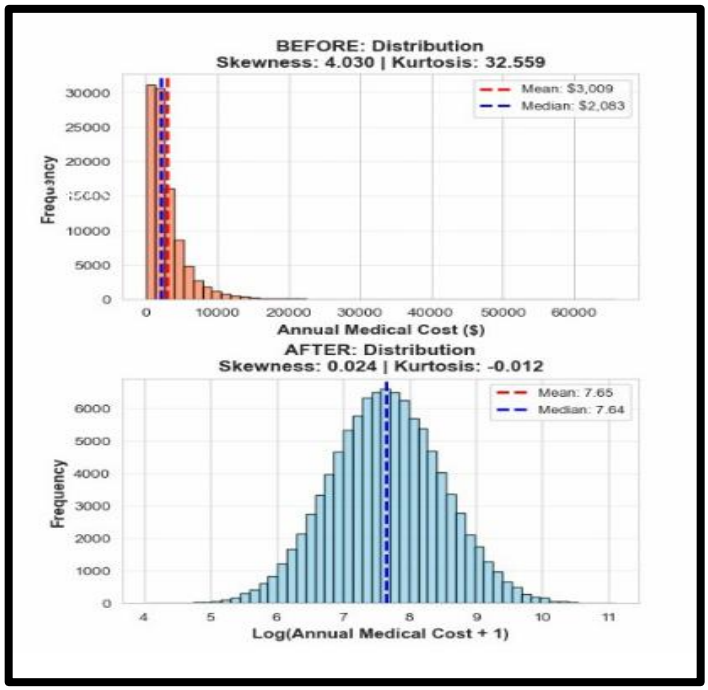
To eliminate noise and redundancy, several features were consolidated or removed:

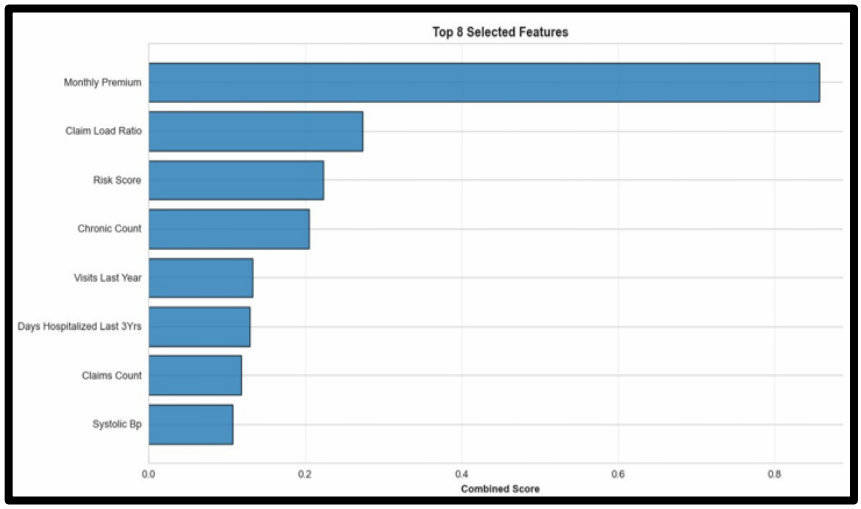
- Multiple illness-related columns were combined into a single chronic_count variable to reduce sparsity and simplify clinical representation
- Redundant financial attributes were dropped due to overlapping information with premium and claim-based variables.
- Various procedure-related columns were merged into total_procedures, offering a more interpretable utilization metric.
- Overlapping or highly correlated risk indicators were removed to prevent multicollinearity and improve model stability.

3.3 Feature Selection

Feature selection was conducted to retain only the most predictive variables. Because the target variable was highly skewed, a log transformation was applied to stabilize variance and improve feature-target relationships. The selection process included:

- Correlation analysis, which highlighted top drivers such as monthly premium, claim_load_ratio, risk_score, and chronic_count.
- Random Forest feature importance, which further confirmed premium, network tier, deductible, risk indicators, and utilization metrics as influential predictors.
- After evaluating 29 candidate variables, a final set of 8 key features (22.9% of total) was selected based on interpretability, predictive value, and consistent ranking across methods.





4. MODEL

4.1 Models Applied

Several supervised regression models were implemented to capture both linear and complex nonlinear relationships in medical cost prediction:

- **Ridge Regression**: A linear model with L2 regularization to reduce overfitting and stabilize coefficients.
- **Neural Network (MLP):** A multi-layer architecture capable of learning nonlinear patterns, using early stopping to prevent overfitting.
- **Random Forest:** An ensemble of decision trees that captures feature interactions and reduces variance through bagging.
- **XGBoost:** A high-performance gradient boosting model optimized for speed and handling complex relationships.
- **CatBoost:** A boosting algorithm that handles categorical features efficiently and mitigates overfitting.
- **LightGBM:** A fast, memory-efficient boosting model designed for large datasets, leveraging leaf-wise growth for high accuracy.

Each model was trained with appropriate parameters based on best practices and initial experimentation.

4.2 Baseline Model Rationale

A diverse set of models was selected to ensure comprehensive evaluation across linear, tree-based, and boosting-based learning approaches:

- **Ridge Regression** served as a baseline due to its interpretability and ability to handle multicollinearity.
- **Random Forest** and **Neural Network** were included to capture nonlinear relationships and interaction effects that linear models may miss.
- **XGBoost, CatBoost and LightGBM** were chosen because boosting models consistently perform well on structured tabular datasets, especially when dealing with heterogeneous features such as demographics, clinical indicators, and healthcare utilization metrics.

Boosting models are particularly suited for this problem because they handle nonlinearity, missing patterns, and interaction effects more effectively than classical methods. The combination of these models ensured a balanced comparison across learning paradigms.

4.3 Hyperparameter Tuning

Hyperparameter tuning was performed to improve model accuracy and generalization.

Why We Chose LightGBM and CatBoost:
We selected LightGBM and CatBoost because they are state-of-the-art gradient boosting algorithms specifically optimized for tabular data, which is the structure of our healthcare dataset.

These models handle non-linear relationships, feature interactions and imbalanced feature distributions significantly better than classical models. Tuning all models would have been computationally expensive and unnecessary. Therefore, efforts were directed toward the two top-performing models.

Why Neural Network Was Not Tuned:
We included a neural network (MLPRegressor) as a baseline to test non-linear modeling capability, but we did not focus hyperparameter tuning on it.

- MLP tuning is computationally expensive
- Tree-based models too showed good accuracy and stability, reducing the value of further neural network optimization.
- Hyperparameter tuning a neural network is significantly more time-consuming than tuning tree-based models such as LightGBM or CatBoost.

To perform tuning, RandomizedSearchCV with 5-fold cross-validation was used. The search explored variations in essential hyperparameters, including the learning rate, number of estimators, maximum tree depth, leaf complexity, subsampling ratios, and regularization strength. The best configuration for each model was selected based on cross-validated R^2, ensuring that the chosen parameters generalize well to unseen data. After tuning, both models showed notable performance improvement, with LightGBM emerging as the best-performing model overall. It achieved the most favorable balance across Test RMSE, Test MAE, and R^2, outperforming all other models in the comparison.

LightGBM Hyperparameter Grid

Hyperparameter	Values Tested	Why This Parameter Matters
n_estimators	200, 400, 600, 800	More trees increase learning capacity; too many cause overfitting. Needed to find the right depth–complexity balance.
learning_rate	0.01, 0.03, 0.05, 0.1	Controls how fast the model learns. Lower rates are stable but need more trees.
num_leaves	15, 31, 63, 127	Main complexity control in LightGBM. Higher values capture more interactions but may overfit.
max_depth	-1, 3, 5, 7	Limits how deep trees grow. Prevents overfitting by restricting tree complexity.
subsample	0.7, 0.8, 1.0	Row sampling for each tree. Improves generalization by reducing overfitting.
colsample_bytree	0.7, 0.8, 1.0	Column (feature) sampling. Reduces tree correlation and overfitting.
reg_alpha	0.0, 1.0, 5.0	L1 regularization. Helps prune unnecessary tree splits and reduce variance.
reg_lambda	0.0, 1.0, 5.0	L2 regularization. Stabilizes leaf weights and prevents overfitting.
min_child_samples	10, 20, 30, 50	Minimum samples per leaf. Larger values avoid overly specific splits (reduces overfitting).

CatBoost Hyperparameter Grid

Hyperparameter	Values Tested	Why This Parameter Matters
iterations	300, 500, 700	Number of boosting rounds. Higher values improve accuracy but risk overfitting.
learning_rate	0.01, 0.03, 0.05, 0.1	Lower rates yield smoother learning and less overfitting; must balance with iterations.
depth	3, 4, 5, 6	Controls tree depth and interaction complexity. Deeper trees learn more patterns but may overfit.
l2_leaf_reg	1, 3, 5, 7, 9	L2 regularization on leaf weights. Crucial for CatBoost to reduce overfitting.
bagging_temperature	0.0, 0.25, 0.5, 1.0	Controls randomness in row sampling. Higher temperature increases diversity among trees.
border_count	32, 64, 128	Number of thresholds for continuous features. Higher values allow finer splits but increase model complexity.

5. MODEL COMPARISON

To evaluate model performance, all trained models were compared using four key metrics: Train Accuracy (R^2), Test Accuracy (R^2), Test RMSE, and Test MAE. These metrics provide a comprehensive view of each model's ability to fit the training data, generalize to unseen data, and minimize prediction errors.

MODEL	TRAIN ACCURACY	TEST ACCURACY	TEST RMSE	TEST MAE
Neural network	0.9596	0.9597	840.00	403.43
LightGBM	0.9611	0.9601	845.92	404.60
CatBoost	0.9593	0.9594	950.05	416.01
XGBoost	0.9499	0.9501	1123.28	468.04
Random Forest	0.9162	0.9134	1707.07	631.82
Ridge Regression	0.6992	0.7009	1500.73	140861.0

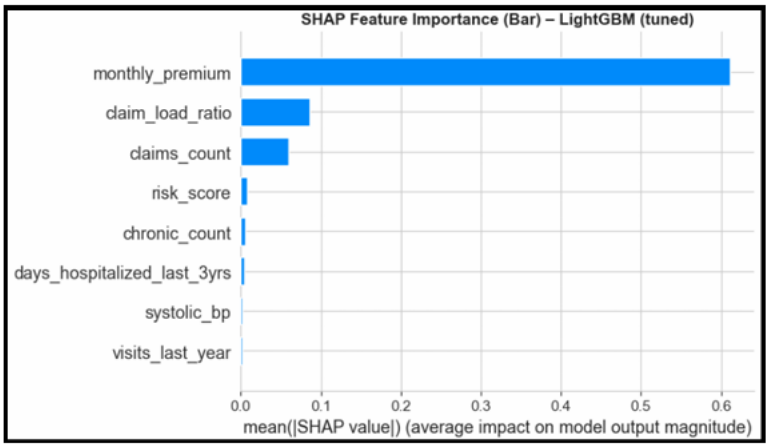
The comparison results show that **LightGBM** and the **Neural Network** achieved the strongest overall performance. LightGBM demonstrated the best balance between bias and variance, achieving the highest Test R^2 (0.9601) and one of the lowest error values. The Neural Network achieved the lowest Test RMSE (840.00), indicating highly accurate predictions, though its MAE was slightly higher than that of LightGBM.

CatBoost also performed competitively, with strong accuracy but slightly higher error metrics compared to the top two models. XGBoost delivered moderate performance, while Random Forest showed signs of underfitting, with noticeable gaps between training and testing results. Ridge Regression performed the weakest due to its inability to capture nonlinear and interaction effects present in the data.

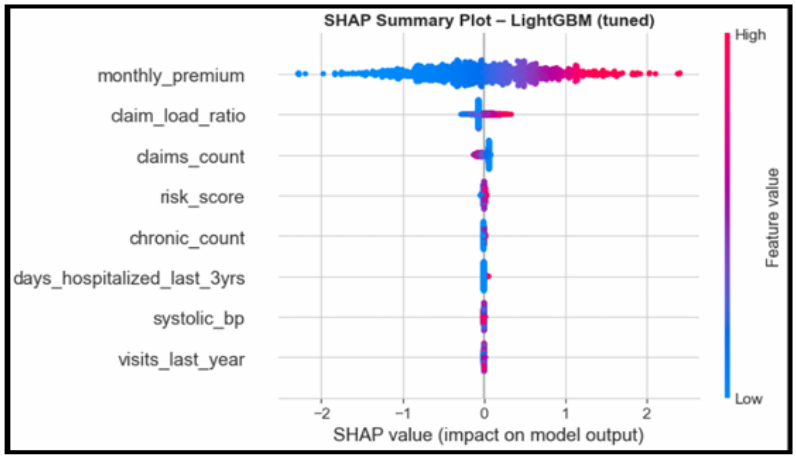
Overall, **LightGBM emerged as the best-performing model**, offering the most consistent accuracy and lowest overall error, followed closely by the Neural Network.

6. SHAP ANALYSIS

To improve interpretability of the final model, SHAP (SHapley Additive exPlanations) values were used to explain how individual features contribute to predicted medical costs. SHAP provides both global features, importance and direction of impact, allowing us to understand not just which features matter most, but *how* they influence predictions.



- Higher monthly premiums strongly push annual medical costs upward, reflecting increased coverage and utilization.
- Claim load ratio and claims count signal financial risk, with higher claim usage pushing predicted medical costs upward.
- Medical risk factors like chronic conditions and past hospitalizations raise costs, but have a smaller impact than financial utilization patterns



- High feature values : (premium, claims, risk scores) consistently push predicted medical costs upward.
- Low feature values: (low premium, low claims, low risk) pull predictions downward, blue dots.
- The plot clearly shows both feature importance and direction of impact, revealing how member behavior and health risk shape cost predictions.

7. LIMITATIONS

Despite strong predictive performance, several limitations should be acknowledged:

- Dependence on Premium-Related Variables - Important predictors such as monthly_premium and claim_load_ratio partially reflect the insurer's pricing strategy, not just medical risk.
- Absence of Time-Based Information - The dataset does not contain year, month, or policy duration information.
- Limited Socioeconomic and Lifestyle Variables - Important external factors income, occupation, exercise, diet, environmental exposures are missing. These significantly influence healthcare costs but are not represented.
- Weak Predictive Signal in Categorical Variables - After encoding, most categorical features contributed little predictive value and were often redundant with numerical ones.

8. CONCLUSION

This project aimed to build a robust machine learning model to predict annual medical costs using demographic, lifestyle, clinical, utilization, and insurance-related features. Through systematic data cleaning, we resolved missing values, removed irrelevant identifiers, and addressed the skewness present in the target variable. Feature engineering further refined the dataset by selecting the most informative predictors and reducing noise, which helped improve overall model performance.

Multiple models were evaluated, and results consistently showed that tree-based boosting algorithms outperformed all other approaches, with LightGBM and CatBoost emerging as the strongest candidates. After applying targeted hyperparameter tuning, LightGBM achieved the best overall performance, delivering the highest predictive accuracy while maintaining low error values.

SHAP analysis provided clear interpretability, revealing that medical costs are primarily influenced by financial utilization metrics such as premium amounts, claim ratios, and claims count, along with clinical indicators like chronic conditions and hospitalization history. These insights validate the model's reliability and align with real-world healthcare cost drivers.

Overall, the project successfully demonstrates that machine learning particularly gradient boosting methods can effectively model and explain medical cost variability. With additional enhancements such as interaction features, extended tuning, and deployment via a lightweight API, this solution can be further extended for real-world insurance pricing and risk assessment applications.